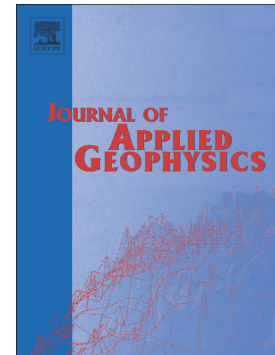


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Combined interpretation of NMR, MICP, and SIP measurements on mud-dominated and grain-dominated carbonate rocks

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ABSTRACT

Carbonate formations still present a challenging target with regard to reservoir characterization as they possess a wide range of pore types and sizes as well as interconnected and isolated pore space. In this study a set of micritic and oomoldic carbonate samples was investigated with three measurement techniques (magnetic resonance (NMR), mercury intrusion (MICP), and spectral induced polarization (SIP, in a frequency range from 5 mHz to 100 Hz) which provide information about a rock's pore space indirectly based on different physical principles. The aims of the study were to combine these pieces of information to characterize the pore space of mud-dominated and grain-dominated carbonate rocks and to investigate potential correlations between pore space properties and SIP-derived parameters.

Pore body and pore throat size distributions were derived from NMR and MICP data, respectively, and interpreted in terms of dominant pore throat (r_t), pore body (r_b) radius and the r_b/r_t ratio. For validation of the NMR-derived r_b distributions, scanning electron microscope images were used for the computation of image-derived pore size distributions. The SIP data were interpreted in terms of characteristic features for oomoldic and micritic samples. Phase shift, real (σ') and imaginary part (σ'') of complex conductivity, respectively, were fitted with Cole-Cole models and the characteristic parameters relaxation time τ , chargeability m , and frequency dependence c were derived for each sample. Subsequently, these parameters were interpreted with regard to correlations with r_b , r_t , and r_b/r_t and specific surface S_{por} .

The results indicate that grain-dominated carbonates possess comparatively large radii r_b and r_t and low S_{por} , which seems to go along with sharp phase peaks (i.e. high c values) and a low absolute phase level (and therefore low chargeability m). The mud-dominated carbonates, in contrast, possess relatively small radii r_b and r_t and high S_{por} which is obviously associated with broad phase peaks (i.e. low c values) and a comparatively high phase level resulting in high m values. τ , in contrast, was found to decrease with increasing ratio r_b/r_t and may therefore have similar values in both micritic and oomoldic samples. Consequently, the results of this study contribute to a better understanding of the petrophysical properties of carbonate rocks as a function of their pore space characteristics.

KEYWORDS

carbonates, pore throat, pore body, nuclear magnetic resonance, spectral induced polarization, mercury intrusion

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INTRODUCTION

Important reservoir parameters such as hydraulic conductivity, storativity or capillary pressure are critically influenced by pore space properties (Mavko et al., 2009; Schön, 2011, 2015). However, many geophysical parameters are not exclusively effected by the absolute amount of pore volume (porosity). Pore space connectivity and constrictivity, representing the significance of either constricted (pore throat) or open (pore body) pore volumes, are the most critical effects (see, e.g., Müller-Huber et al., 2015; Müller-Huber et al., 2016a, 2016b). While the pore space of sandstone reservoirs is rather homogeneous and dominated by particle characteristics such as sorting, shape or rounding, carbonate rocks possess a wide range of pore types, pore sizes and pore interconnection types. This results from the fact that they are deposited and lithified in various depositional environments influencing mineralogy, grain morphology, and post-depositional diagenetic processes (Choquette and Pray, 1970).

The relaxation process investigated with magnetic resonance (NMR) measurements exploits the fact that many nuclei such as hydrogen nuclei of water or hydrocarbons possess an inherent magnetic moment and angular momentum which provide them with the ability to interact with a magnetic field. Thus NMR measurements are sensitive to the type and properties of the pore-filling fluids, their interactions in rock pores, and surface relaxivity and may therefore provide information about pore body size distribution (e.g. Coates et al., 1999; Dunn et al., 2002). Capillary forces in reservoir rocks are determined by the surface and interfacial liquid tensions, pore-throat size and shape and the wetting properties of the rock matrix. Therefore, studying the intrusion of a non-wetting phase (mercury) being injected into a rock sample at increasing pressures during mercury intrusion measurements (MICP) provides pore throat dimensions which can be derived with the Washburn equation (Washburn, 1921) and an area-equivalent diameter of the throats connecting the pore system (Basan et al., 1997). Spectral induced polarization (SIP) measurements in the low-frequency range (below 1 kHz) investigate the fluid-filled pore volumes, being sensitive to electrochemical processes of electronic-ionic interaction, interface properties of the grain-fluid contact region, cation-exchange phenomena, or pore constrictivity (Börner, 2009; Schön, 2015). Thus the critical influence of specific surface is well established (Börner and Schön, 1991; Börner, 1992; Weller et al., 2015; Weller et al., 2016), while a potential interconnection between peak frequencies and pore space parameters has also been discussed recently (e.g. Lesmes and Morgan, 2001; Scott and Barker, 2003; Leroy et al., 2008; Kruschwitz et al., 2016). Lesmes and Frye (2001) proved the influence of pore fluid chemistry on IP parameters (quadrature conductivity, metal factor, normalized chargeability) of sandstone.

In carbonate rocks, the NMR measurement is a common laboratory and/or logging method used to estimate, for instance, porosity, pore size distribution, irreducible water saturation, producible fluids, and permeability using prediction methods established for sandstone reservoirs (e.g. Allen et al., 2000; Allen et al., 2001) which include additional parameters considering the carbonates' more complex pore space geometry and pore interconnection types (e.g. Chang et al., 1994; Amabeoku et al., 2001; Hidajat et al., 2004; Westphal et al., 2005; Di Rosa et al., 2006; Chen et al., 2008). Recently, more sophisticated NMR interpretation approaches for carbonate rocks deal with wettability characterization (Looyestijn, 2007; Kesserwan et al., 2016), studying porosity and (residual) fluid saturation under reservoir conditions (Liu et al., 2015) or in combination with

resistivity-derived fluid saturations (Thern, 2014; Thern and Page, 2016), or improved derivation of surface relaxivity (Mesquita et al., 2016) and fluid typing (Chen et al., 2016). With regard to pore space characterization in carbonate rocks, approaches for an improved derivation of capillary pressure curves and pore size distributions from NMR measurements (Salimifard et al., 2015) and a combined interpretation of NMR-derived pore space characteristics with direct methods such as X-ray microtomography (Denisenko et al., 2015) have been presented recently. The authors conclude that in order to cope with the wide scale range of carbonate pores, a combination of different measurement methods and thus different resolutions is needed.

As complex conductivity measurements provide an estimate of the specific surface (e.g. Börner and Schön, 1991; Börner et al., 1996), they are typically used to derive, for example, geohydraulic parameters such as hydraulic conductivity (Slater and Lesmes, 2002; Revil and Florschütz, 2010) or environmentally relevant information about potential contaminants and their distribution in the pore space (e.g. Olhoeft, 1985; Börner et al., 1993). So far, the complex conductivity behavior of sandstones, building materials and sediment samples has been studied quite intensely (e.g., Tong et al., 2006; Titov et al., 2010; Kruschwitz et al., 2010; Börner and Müller-Huber, 2016). However, only few studies deal with frequency-dependent electrical resistivity and excess conductivity in carbonate rocks. Excess conductivity in carbonate rocks is typically attributed to clay minerals. Nevertheless, Saner and Akbar (2010) verified the significance of excess electrical conductance by multiple salinity tests in clay-free carbonates, especially in carbonates with high amounts of micrite which increases specific surface and microporosity. In macroporous carbonates, however, they only found marginal contributions of excess conductance. Cerepi (2004) investigated frequency-dependent electrical resistivity in carbonate rocks and calculated a chargeability factor. The author found that chargeability is highest in mudstones-wackestones and decreases with increasing amounts of vuggy or karst-associated pores. Halisch et al. (2014) differentiated four different phase spectra types in Egyptian carbonate samples of similar fabric types with comparably low micrite content (packstones, grainstones, floatstones). Norbistrath (2016) investigated dolomite samples and found that phase magnitudes increase with decreasing porosity. The complex conductivity of calcite was studied by Wu et al. (2010) who proved that the polarization of the electrical double layer can be observed in calcite, even though the mineral phase is non-conductive. More recently Li et al. (2016) studied the apparent zeta potential of calcite and found the latter to be strongly influenced by surface conductivity effects.

With regard to these findings, it seems likely that a combination of NMR, MICP, and SIP techniques, which all investigate pore space features but are based on different physical principles, may deliver complementary information. The objectives of this study are, therefore,

- 1) to combine the three different measurement techniques in order to enhance our knowledge about carbonate pore space by using the partly complementary information provided by each individual technique and
- 2) to explore potential correlations between specific pore space properties such as pore body and pore throat size and SIP parameters to gain deeper insight into the fundamental parameters influencing SIP measurements in carbonate rocks with complex pore systems.

To achieve these aims, a set of ten mud-dominated and grain-dominated carbonate samples was selected for investigations and a comprehensive measurement program was performed. NMR pore body radius and MICP

pore throat radius distributions were derived and are presented along with the SIP spectra. All pieces of information are then combined to derive characteristic features of the studied carbonate samples and interpreted in terms of potential correlations with relaxation time τ , chargeability m , and frequency dependence c . To verify the validity of the NMR-derived pore radius distributions, pore size distributions based on selected scanning electron microscope (SEM) images and image processing techniques are used.

THEORY

For fluids filling a rock's pore space, three independent NMR relaxation mechanisms are involved: Bulk fluid processes, surface relaxation, and diffusion in the presence of magnetic field gradients (e.g., Coates et al., 1999; Dunn et al., 2002). The T_2 surface relaxation term, $T_{2,surf}$ [s], is given in equation (1) where ρ_2 [$\mu\text{m/s}$] is the T_2 surface relaxivity and S_{por} [μm^{-1}] is the pore surface to pore volume ratio (e.g. Brownstein & Tarr, 1979; Kleinberg et al., 1994; Kenyon, 1988).

$$\frac{1}{T_{2,surf}} = \rho_2 S_{por} \quad (1)$$

For simple pore shapes, S_{por} can be related to pore body radius (e.g., Marschall et al., 1995; Schön, 2011; Schön, 2015) and is thus a measure of pore size (Coates et al., 1999). S_{por} equals $3/r$ for spherically shaped pores and $2/r$ for cylindrically shaped pores. Surface relaxation occurs at the fluid-solid interface, i.e. the grain surface of rocks. It is a measure of a surface's ability to cause protons to relax or to lose their phase coherence due to magnetic interactions at the fluid-solid interface.

To estimate surface relaxivity, the method presented by Katika et al. (2014) was used in this study. The described approach is based on the assumption that at the maximum T_2 relaxation time, BET- and NMR-derived specific surface areas are equal. The BET specific surface S_{BET} in [m^2/g] can be transformed into the pore-surface-to-pore-volume ratio S_{por} using matrix grain density ρ_g and porosity Φ (Pape et al. 1982, 1984, 2006)

$$S_{por} = S_{BET} \cdot \rho_g \frac{(1 - \Phi)}{\Phi} = \frac{S_{pore}}{V_{pore}} = \frac{S_{tot}}{\Phi} \quad (2)$$

where S_{tot} [μm^{-1}] is the pore surface area normalized to total sample volume.

Mercury intrusion (MICP) is controlled by the sizes of pore throats (Marschall et al., 1995). Consequently, MICP provides a pore throat size distribution by means of the Washburn (1921) equation:

$$r_c = \frac{-2 \gamma \cos \theta}{p_c} \quad (3)$$

where p_c [psi or Pa] is capillary pressure, γ [dynes/cm² or mN/m²] is the surface tension of mercury, θ [°] is its contact angle in air and r_c is the critical pore throat radius [μm] for a cylindrical pore geometry. Typical values for contact angle and air-mercury interfacial tension are 140 degrees and 480 dynes/cm², respectively (see e.g. Adamson and Gast, 1997). The critical pore throat values represent the radii at which maximum intrusion of the nonwetting phase occurs at a comparatively minor pressure increase.

Induced polarization is a current-stimulated electrical phenomenon observed as a delayed voltage response in rocks or soils resulting from a capacitive behavior (Ward, 1990), which can be studied in the time or frequency domain. It is caused by the specific electrochemical structure of the fluid-rock interface, which exhibits a charge separation at the rock surface (the so-called electrical double layer). As a result the rock reacts with both electric conduction and polarization to an external electric field. Therefore, electrical resistivity and electrical conductivity, respectively, are frequency-dependent properties with a real (σ') and an imaginary component (σ''), representing ohmic conduction and polarization mechanisms. They are related to the real-valued electrolytic conductivity in the pore water σ_{el} and the complex-valued interface conductivity (caused by the electrical double layer) σ_{int}^* (Vinegar and Waxman, 1984; Schön, 2015):

$$\sigma^*(\omega) = \sigma' + i\sigma'' = \sigma_{el} + \sigma_{int}^* \quad (4)$$

The magnitude of conductivity $|\sigma|$,

$$|\sigma| = \sqrt{(\sigma')^2 + (\sigma'')^2} \quad (5)$$

as well as the phase shift (φ),

$$\varphi = \arctan \left[\frac{\sigma''}{\sigma'} \right] \quad (6)$$

are determined by applying a sinusoidal voltage between two current injection electrodes and measuring the sinusoidal voltage due to the resulting current flow at two potential electrodes (e.g., Olhoeft, 1985).

A frequently used empirical model to fit complex electrical resistivity data in consolidated rocks is the Cole-Cole model (Cole and Cole, 1941) which was also used by Pelton et al. (1978) in ore exploration. The Cole-Cole model describes a resistive network, which is composed of two resistances and a complex impedance term and represents the model conception of a rock harboring both free, water-filled pore paths and paths blocked by polarizable particles. Nowadays, the model is frequently used to describe various types of polarization in rocks. Pelton et al. (1978) introduced the following expression for complex impedance $Z(\omega)=1/\sigma^*(\omega)$

$$Z(\omega) = R_0 \left[1 - m \left(1 - \frac{1}{1 + (i\omega\tau)^c} \right) \right] \quad (7)$$

where R_0 [Ohmm] is DC resistivity, ω [Hz] is the angular frequency of the polarizing current, m [-] is the chargeability (ratio of voltage after, to the voltage before cessation of an infinitely long charging current), τ [s] is the relaxation time constant, and c [-] is an exponent describing the slope of the phase spectrum at a double logarithmic plot.

A range of theories exists to describe the physical origin of individual relaxation phenomena that add up to the measured SIP spectra in ore-free rocks. A review of physical pore-scale theories can be found in Kruschwitz et al. (2016). One approach states that electrolytic interface polarization results from different ion mobility and diffusion processes causing a charge buildup in nonmetallically mineralized rocks. Membrane polarization in porous rocks is such a phenomenon (Marshall and Madden, 1959; Buchheim and Irmer, 1979; Vinegar and Waxman, 1984) which can be caused either by negatively charged clay particles accumulating cations thus forming a barrier for anions, or by constrictions of the pore space. Recently the dependency of relaxation time

on characteristic length parameters of the pore space (e.g., pore throat size) has been discussed (see, for example, Titov et al., 2002; Scott and Barker, 2003; Volkmann and Klitzsch, 2010; Revil et al., 2012; Bückner and Hördt, 2013a; 2013b). Other studies (see e.g. Lima and Sharma, 1992; Revil and Florsch, 2010) established the relationship between chargeability m and charge concentration and mobility, between exponent c and the distribution of grain sizes and between relaxation time constant τ and grain radius and properties of the electrolyte in granular media and unconsolidated shaly sands.

To date, however, no generally applicable relationship between relaxation time and dominant pore throat size has been found for consolidated carbonate rocks.

MATERIALS AND METHODS

A set of Lower Muschelkalk samples (Middle Triassic, Anisian) was selected for petrophysical investigations. At the sampling site Rüdersdorf near Berlin (Germany) located in the central part of the Germanic (or Central European) Basin, the Lower Muschelkalk is typically composed of two facies types, which represent different energetic conditions on a carbonate ramp: Wellenkalk is composed of successions of fine-grained, dark grey marly peloidal/micritic limestones and marl of stillwater environments (Fig. 1, left) alternating with biogenic/peloidal shell limestones or tempestites. Characteristic features are undulating compaction structures caused by alternating deposition of sediments of different densities (Zwenger, 1993). The yellowish oomoldic limestones of the Schaumkalk (Fig. 1, right) were deposited under high-energetic, shallow marine conditions. They are characterized by partly cemented, but mostly open spherical to ellipsoidal pores which were formed by dissolution of ooids (Bodzioch and Kowal, 2001; Noack and Schroeder, 2003). Hardgrounds with signs of bioerosion and occasional micritic limestone layers may be present as well (Zwenger and Koszinski, 2009).

NMR and SIP measurements were conducted on cylindrical plugs of 5.5 cm length and 3.5 cm diameter which were drilled from samples collected at an open pit mine. All plugs were dried until a constant mass was measured and subsequently saturated with brine. To avoid dissolution of the carbonate samples, 1 L distilled water was mixed with 1 g dolomite, 1 g limestone, and 1 g NaCl, allowed to rest until a constant brine conductivity value could be registered and filtrated afterwards (electrical conductivity: 3.5 mS/cm at 19°C).

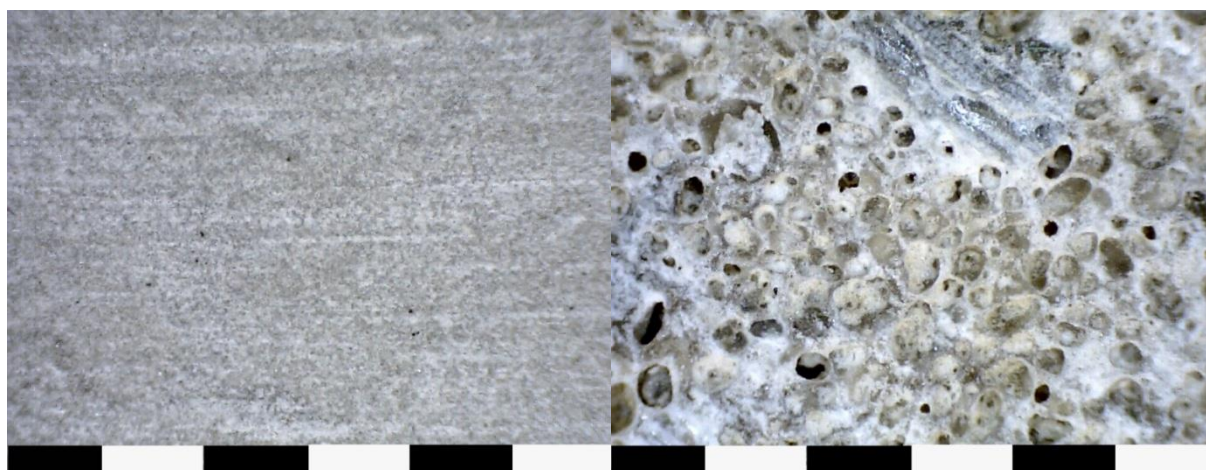


Fig. 1 – Photomicrographs of a mud-dominated micritic Wellenkalk (left) and a grain-dominated oomoldic Schaumkalk sample (right). Scale bar: mm.

All NMR measurements were performed using a Maran Ultra 2 MHz spectrometer (Resonance Instruments). To avoid desiccation, the saturated carbonate samples were sealed with Teflon strap and inserted into the NMR apparatus. The transverse magnetization decay curve was measured using a Carr-Purcell-Meiboom-Gill sequence (CPMG; Carr and Purcell, 1954; Meiboom and Gill, 1958). At least 128 scans were acquired to ensure signal-to-noise ratios greater than 100. A multi-exponential relaxation model was used to invert the echo trains into transverse relaxation (T_2) distributions.

For the SIP measurements, a SIP-Fuchs II system (Radic Research), which measures complex resistivity over 7 decades of frequency (1.4 MHz up to 12 kHz) was used. The rock samples were implemented in a measuring cell, which realized the measurement by means of a 4-electrode configuration. Thereby, lead plates served as current electrodes and pure platinum wire electrodes, which were positioned in narrow slits well away from the high current density, were used as potential electrodes. The measurements start with the highest frequency f and subsequently n further frequencies are measured according to the relationship $f/2n$. To avoid current bypassing the samples, Teflon strap was used to seal them in the measurement cell. Measurements were carried out at an air-conditioned laboratory where a constant temperature of 19°C was ensured to avoid potentially occurring temperature effects. All samples were equilibrated to that temperature before the measurements.

Mercury intrusion measurements were carried out according to DIN 66133 using the Quantachrome PoreMaster 60-GT apparatus which generates pressures up to 60,000 psi for pore size analysis of pores ranging from larger than 950 μm to as small as 3 nm in diameter. Using this equipment the pore throat size distributions and dominating pore throat radii were determined for all samples.

To enhance the insight into pore space properties provided by the methods mentioned above, BET gas adsorption measurements were carried out. The specific (mass-related) surface area S_{BET} was measured with a surface area analyzer (Gemini 2360, Micromeritics) using the standard multipoint BET method employing nitrogen adsorption (e.g. Brunauer et al., 1938). The total solid surface can be measured with the standard BET method from the condensation of a gas at a solid surface given that the surface occupied by one adsorbed

molecule is known (e.g. Vincent et al., 2011). The dried end pieces left after cutting and trimming the plugs were used for BET experiments to minimize potential heterogeneity effects. Table 1 gives an overview of S_{por} calculated from S_{BET} , matrix grain density ρ_g and porosity Φ of the carbonate samples.

Table 1: Pore space parameters of the studied carbonate samples.

Sample	Rock Type	S_{por} [μm^{-1}]	S_{BET} [m^2/g]	Φ [-]	Sample	Rock Type	S_{por} [μm^{-1}]	S_{BET} [m^2/g]	Φ [-]
R 2/1	marly/micritic Wellenkalk	43.9	1.126	0.065	R 14/1	oomoldic Schaumkalk	3.4	0.334	0.228
R 3/1	marly/micritic Wellenkalk	86.9	2.142	0.065	R 15/2	oomoldic Schaumkalk	4.4	0.376	0.215
R 4/1	marly/micritic Wellenkalk	128.2	2.843	0.054	R 16/1	oomoldic Schaumkalk	4.2	0.522	0.270
R 7/1	micritic Schaumkalk	18.2	1.742	0.211	R 20/2	oomoldic Schaumkalk	3.5	0.596	0.343
R 11/2	micritic Schaumkalk	19.4	0.708	0.086	R 21/1	oomoldic Schaumkalk	5.0	0.879	0.332

Using the method from Katika et al. (2014), surface relaxivities between 0.1 and 21 $\mu\text{m}/\text{s}$ were calculated for the investigated samples. These ranges correlate well with the findings of other authors who report values between 3-8 $\mu\text{m}/\text{s}$ for grain-supported and about 1 $\mu\text{m}/\text{s}$ for mud-supported fabrics (Quintero et al., 1999). Bhagat and Sondergeld (2012) measured relaxivities as low as 0.1 $\mu\text{m}/\text{s}$ up to 14 $\mu\text{m}/\text{s}$ for UAE carbonates. These observations highlight that each carbonate formation type is associated with a characteristic surface relaxivity and selecting an average value may not be representative when, for example, a large depth section is logged.

Additionally, SEM images in secondary electron mode of unpolished carbon-coated pieces of selected samples were used for the derivation of image-based pore body size distributions. Images were generated with a Hitachi S-2700 microscope at magnifications of 22 to 10,000 times. The necessary image processing was carried out with the open-source software package ImageJ which enables particle measurements by application of a threshold. The processing steps of this image processing routine for the raw images were as follows: Setting the scale to ensure correctly scaled particles and pores, applying a FFT bandpass filter that filters structures according to their size, using the contrast enhancement tool of ImageJ, adjusting pixel saturation and mean filtering by a mean of 1 to 3 pixels which was selected depending on whether the studied sample was dominated by small or large pores. Afterwards, representative sections of the images and an appropriate threshold level were selected which ensured correctly sized pores visible as dark spots in the image on the one hand and allowed for a reliable differentiation between pores and solid rock matrix without leading to overweighting of individual images when computing the pore size distribution on the other hand. Finally, the pores' area, their minimum and maximum Feret diameter and its angle (Merkus, 2009) were determined. As suggested by Merkus (2009) the maximum Feret diameter is the distance between two parallel tangents on opposite sides of the image of a randomly oriented particle while the minimum Feret diameter is the distance between two parallel tangents which are parallel to the maximum Feret diameter in its position of maximum

stability. The Feret angle determines the angle between the maximum Feret diameter and the vertical line from the projection. For small-scale images (500x), pores exceeding a cross sectional area of $1 \mu\text{m}^2$ could be detected, while pores larger than $200\text{--}300 \mu\text{m}^2$ were studied in large-scale images (22x).

EXPERIMENTAL RESULTS AND DISCUSSION

In order to study carbonate pore space characteristics of the micritic and oomoldic samples in detail and to clarify the influence of pore space dimensions on petrophysical properties, the results of NMR, MICP, and SIP measurements on 10 selected carbonate samples are presented and discussed in the following section.

The left-hand side of Fig. 2 presents the result of NMR T_2 measurements in terms of NMR-derived pore body size distributions. The distributions were calculated assuming a cylindrical pore shape for the marly/micritic Wellenkalk samples where no pores except for a few microfractures can be observed under magnification. A cylindrical pore shape approximation was used in this case as fractures tend to have greater length compared to their aperture so that they are more similar to cylinders than spheres with regard to geometry. Spherical pore shape for the oomoldic and micritic Schaumkalk samples where pores with a more or less circular cross section can be observed with the naked eye (c.f. Fig. 1). The different behavior of mud-supported and grain-supported carbonates is quite obvious: All micritic Wellenkalk samples are associated with much smaller pores of typically lower than $0.01 \mu\text{m}$ in radius and a very limited amount of pores with a radius of 0.1 up to $1 \mu\text{m}$. All oomoldic samples are characterized by pore body radii of about 0.1 to larger than $100 \mu\text{m}$. Sample R 15/2 represents, however, an exception as a limited amount of very small pores with a radius as low as $0.01 \mu\text{m}$ can be found.

As probably expected, the micritic Schaumkalk samples occupy an intermediate position with pore body radii ranging from as low as 0.001 and $0.1 \mu\text{m}$, respectively, up to larger than $1 \mu\text{m}$. It is, however, necessary to keep in mind that the distributions displayed in Fig. 2 represent the whole spectrum of T_2 relaxation times including clay-bound water which does not represent pore space. A T_2 cutoff of 3 ms for clay-bound water (e.g., Coates et al., 1999; Allen et al., 2000; Dunn et al., 2002; Schön, 2011) results in a pore radius cutoff of $0.001\text{--}0.0026 \mu\text{m}$ for marly/micritic Wellenkalk, $0.01\text{--}0.03 \mu\text{m}$ for micritic Schaumkalk samples and $0.05\text{--}0.2 \mu\text{m}$ for the oomoldic Schaumkalk samples. All features larger in size could thus be considered to be non-clay-related.

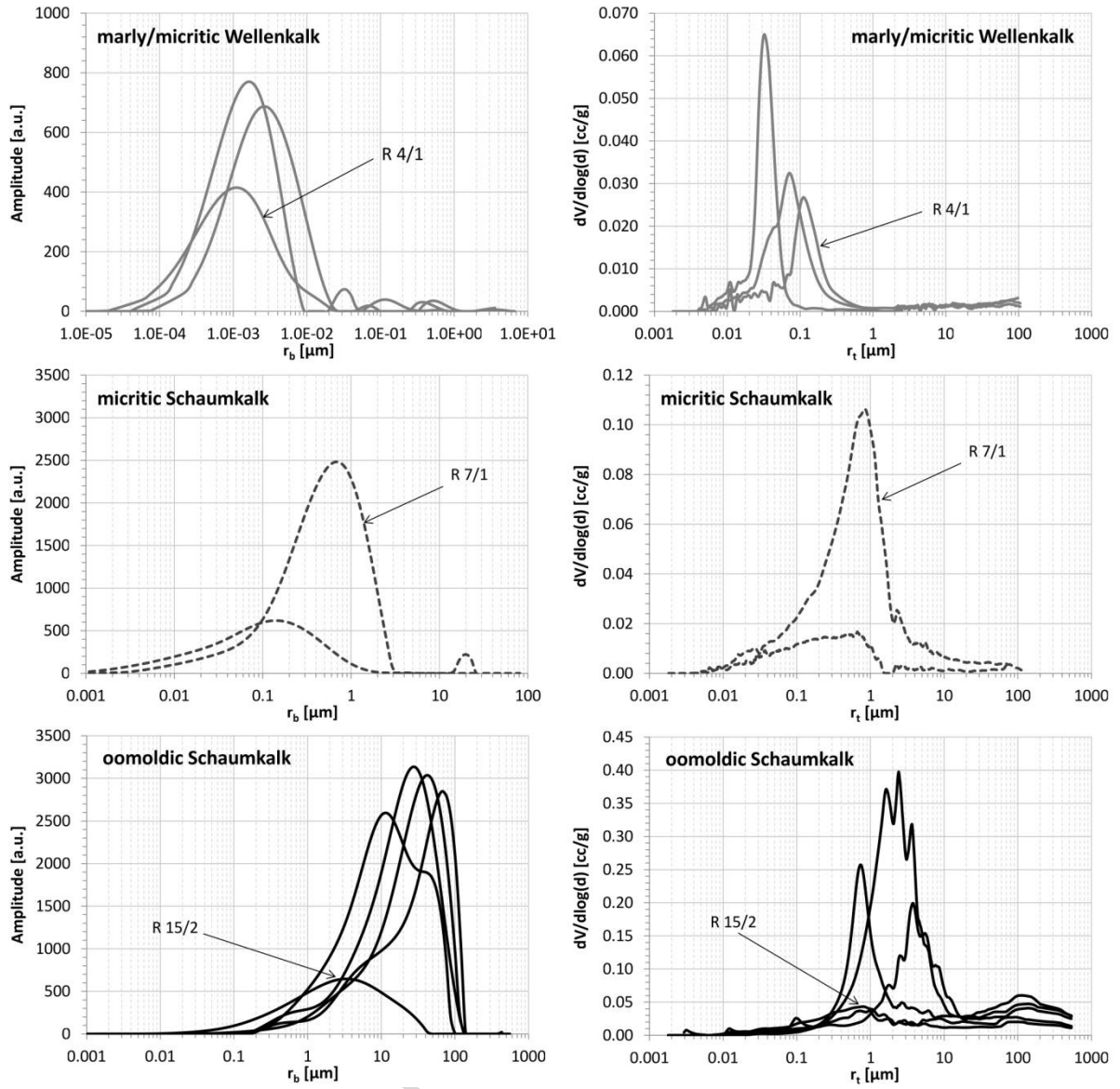


Fig. 2 – Left: NMR pore body radius distributions; right: MICP pore throat radius distributions. Marly/micritic Wellenkalk: solid gray lines; micritic Schaumkalk: stippled lines; oomoldic Schaumkalk: solid black lines.

The right-hand side of Fig. 2 presents the MICP-derived pore throat size distributions. Again, mud-supported and grain-supported samples possess different characteristics: All oomoldic Schaumkalk samples are associated with pore throat radii ranging predominantly from 0.1 to 10 μm and a limited amount of pore throats larger than 100 μm. In contrast, the micritic Wellenkalk samples are associated with much smaller pore throats of typically lower than 1 μm in radius to as low as about 0.005 μm. As far as pore throat radii are concerned, the micritic Schaumkalk samples are associated with similar characteristics as the oomoldic ones with pore throat radii of predominantly 0.01 to 10 μm.

Marschall et al. (1995) stated that a direct matching of NMR and MICP data on a common scale is only possible, if the shape of both distributions is similar and a consistent pore body-to-throat size ratio exists. In this case, a T_2 distribution could be computed from MICP data using effective relaxivity ρ_e , which is proportional to the

product of the intrinsic relaxivity ρ_2 and a pore throat-to-body size ratio and assuming, for instance, a cylindrical pore geometry. This concept was further elaborated by, for instance, Mesquita et al. (2016).

For the studied Muschelkalk samples, the following observation can be made with regard to the correlation between pore body and pore throat radii: For most oomoldic samples, pore throat radii are one to two orders of magnitude smaller than their pore body radii (Fig. 3). Only sample R 15/2 presents an exception as its pore body radii are larger than its pore throat radii only by a factor of about 5.

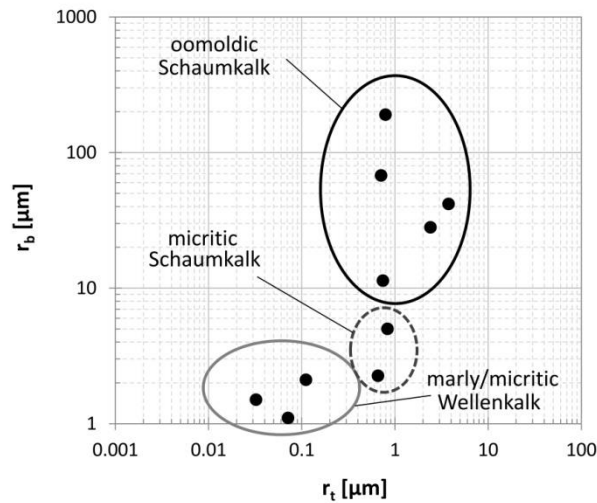


Fig. 3 – NMR- or SEM-derived dominant pore body radius r_b [μm] vs. MICP-derived dominant pore throat radius r_t [μm].

The NMR and MICP pore size distributions of both the micritic Schaumkalk and the micritic Wellenkalk samples possess a similar shape but as far as the absolute values of the radii are concerned, the NMR-derived radii of the Wellenkalk samples seem to be much smaller. It does not seem likely that such large volumes of pores are below the MICP detection limit. This bias in the NMR-derived distributions may be caused by small amounts of pyrite in the samples which could be observed in some SEM images and affect the T_2 relaxation. Framboidal pyrite may occur in burrowed, pelitic limestone layers of the Wellenkalk caused by early diagenetic, bacterially induced sediment mineralization and anaerobic conditions (Bodzioch and Kowal, 2001). Surface relaxivity is increased by iron-bearing minerals so that relaxation times are shifted towards shorter T_2 times (e.g. Keating and Knight, 2007; 2010). This means that the BET-based surface relaxivities, which only reflect the relaxing strength of the grain surfaces, are too low and thus pore body radii are calculated too small.

For validation of the NMR-derived pores size distributions, they were compared to pore size distributions computed from SEM image analysis (Fig. 4). For computation of pore size distribution based on SEM images, a few general assumptions were made: The investigated samples were assumed to be isotropic and homogeneous and the minimum Feret diameter was selected for representation of the pore diameter to avoid overestimation of pore size in case of connected pore systems. Image porosity was defined as the sum of all pore cross sectional areas divided by the total image area.

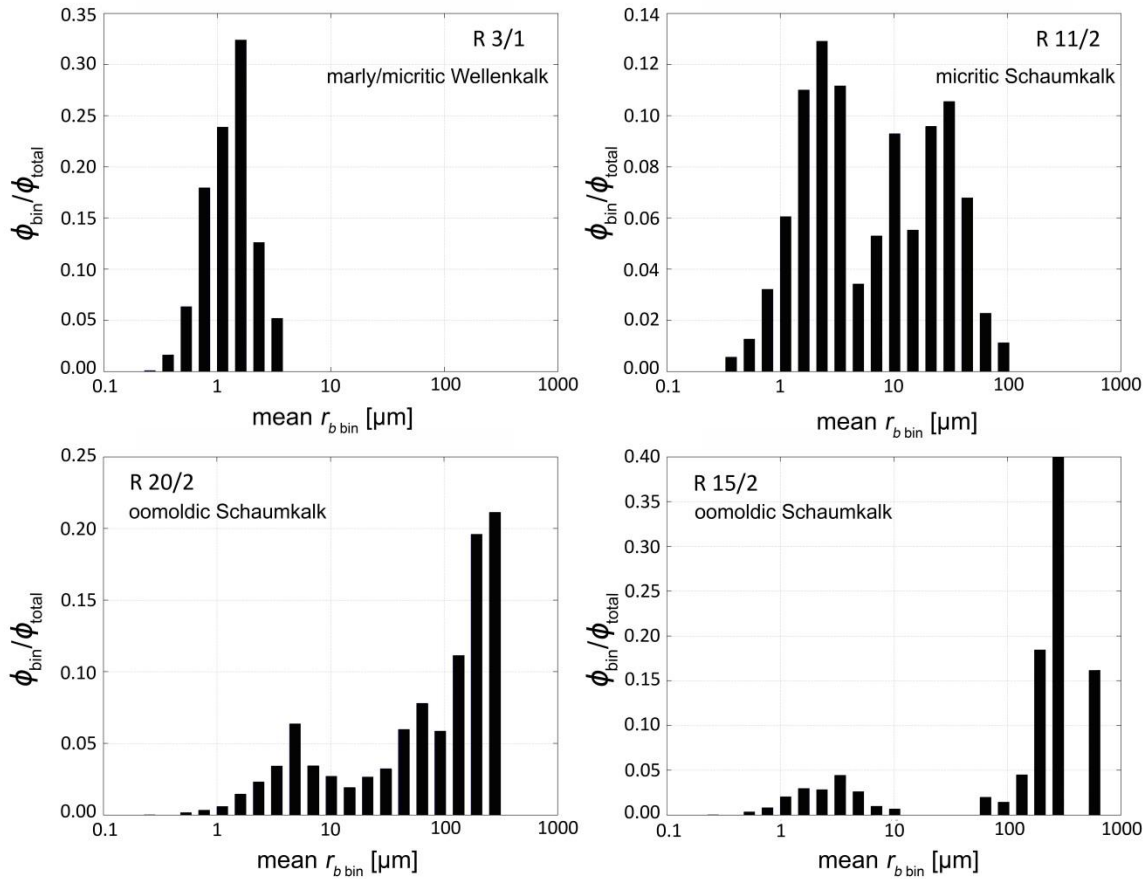


Fig. 4 – Examples of SEM image-derived pore body radius r_b distributions of a marly/micritic Wellenkalk (top left), a micritic Schaumkalk (top, right) and two oomoldic Schaumkalk samples (bottom).

For all images considered for one sample, the pore radius was calculated as half of the pore diameter, the area fraction each pore was adding to the total pore area of an image was calculated and image porosity was approximated as the sum of each image's pore area divided by the image's total area. Total image porosity was then calculated from several images multiplied by weighting factors in order to add up to the measured total porosity. This means that image porosities from different images can be summed up by applying a weighting factor for each individual image's porosity to sum up to directly measured total porosity. As the largest pores add most significantly to porosity, the large-scale images covering them most reliably received the highest weighting factor of 1. Finally, for the computation of pore radius distributions, pore radius bins were defined and their contributions to total porosity were calculated.

A comparison of NMR- and image-derived pores size distributions highlights that each method has its specific capacity to resolve pore sizes. While SEM can resolve even large pores with a radius of over 500 μm (Fig. 4), these largest pores cannot be detected by NMR measurements. The likely cause of this shortcoming is that despite wrapping them with Teflon strap, it is difficult to keep the largest pores fully saturated. If the pores are only partially water-saturated or even water-free, they cannot be detected in their original size. This seems to apply especially to the oomoldic sample R 15/2 with its large pores that have obviously been created by the

dissolution of fossil fragments. The largest pores with a radius of about 500 μm could not be resolved by NMR measurements at all (Fig. 4).

With regard to image-derived pore size distributions, the smallest pores that can be resolved have a radius of about 0.5 μm . This is obvious since the resolution capacity was given with pore cross sectional areas exceeding 1 μm^2 for the smallest images, which results in a pore radius of 0.56 μm if a circular cross sectional shape is assumed. Nevertheless, the SEM image-derived pore size distributions of the micritic samples seem much more plausible than the NMR-derived ones, especially in comparison with MICP pore throat size distributions. Thus the dominant pore body sizes computed from SEM image analysis were used for further considerations (Fig. 3, Table 2) for all micritic samples.

Based on the investigations of pore space characteristics the measured SIP spectra may be interpreted. In Fig. 5 we concentrate on the frequency dependence of the phase, because phase peaks are frequently believed to correlate with characteristic pore space features. In addition, real part σ' and imaginary part σ'' of complex conductivity are also shown in Fig. 5. A frequency range up to 100 Hz was studied to avoid potential bias of measured data due to coupling effects. Generally the absolute values of phase response of the studied carbonate samples are a bit lower with a phase maximum of 10-12 mrad as compared to 10 up to 25 mrad measured by Halisch et al. (2014). The imaginary part σ'' of complex conductivity was found to be highest for samples with high micrite content (marly/micritic Wellenkalk, micritic Schaumkalk). The micritic samples are characterized by $3.0 \cdot 10^{-3} < \sigma' < 1.0 \cdot 10^{-2}$ [S/m] and $1.0 \cdot 10^{-5} < \sigma'' < 1.5 \cdot 10^{-4}$ [S/m].

The oomoldic Schaumkalk samples, in contrast, are characterized by low absolute phase values of generally less than 5 mrad and plot at $2.0 \cdot 10^{-3} < \sigma' < 7.0 \cdot 10^{-3}$ [S/m] and $\sigma'' < 2.0 \cdot 10^{-5}$ [S/m]. These observations clearly support the earlier findings of Cerepi (2004) or Saner and Akbar (2010) who suggested that micrite content is a critical factor increasing excess electrical conductivity due to, for instance, increased specific surface. This is the case for the studied samples where specific surface is highest for micritic carbonates (Table 1).

Most marly/micritic Wellenkalk and oomoldic Schaumkalk samples except for R 4/1 (plateau between 0.5 and 50 Hz) and R 15/2 (no peak or plateau) possess a more or less prominent phase peak at a frequency of about 23 Hz, whereas the micritic Schaumkalk samples exhibit a phase peak at about 3 and 12 Hz, respectively. As NMR T_2 distributions indicate significant amounts of clay minerals especially in the micritic Wellenkalk samples, their effect on SIP measurements may have to be considered in addition to the effect of pore space geometry. The NMR T_2 distributions indicate that all oomoldic samples except for R 15/2 are clay-free, which suggests that the slight peaks that can be observed in the SIP phase spectrum are not related to clay minerals but rather to pore space features such as a limited amount of small-sized pores with a radius of 0.1-1 μm . This conclusion is supported by the characteristics of sample R 15/2 which possesses the highest clay content among the oomoldic samples but shows no peak in the phase spectrum at all. Thus it seems likely that these peaks are not related to clay minerals. Similar observations can be made for the micritic Schaumkalk samples which have only minor clay content but exhibit a prominent phase peak.

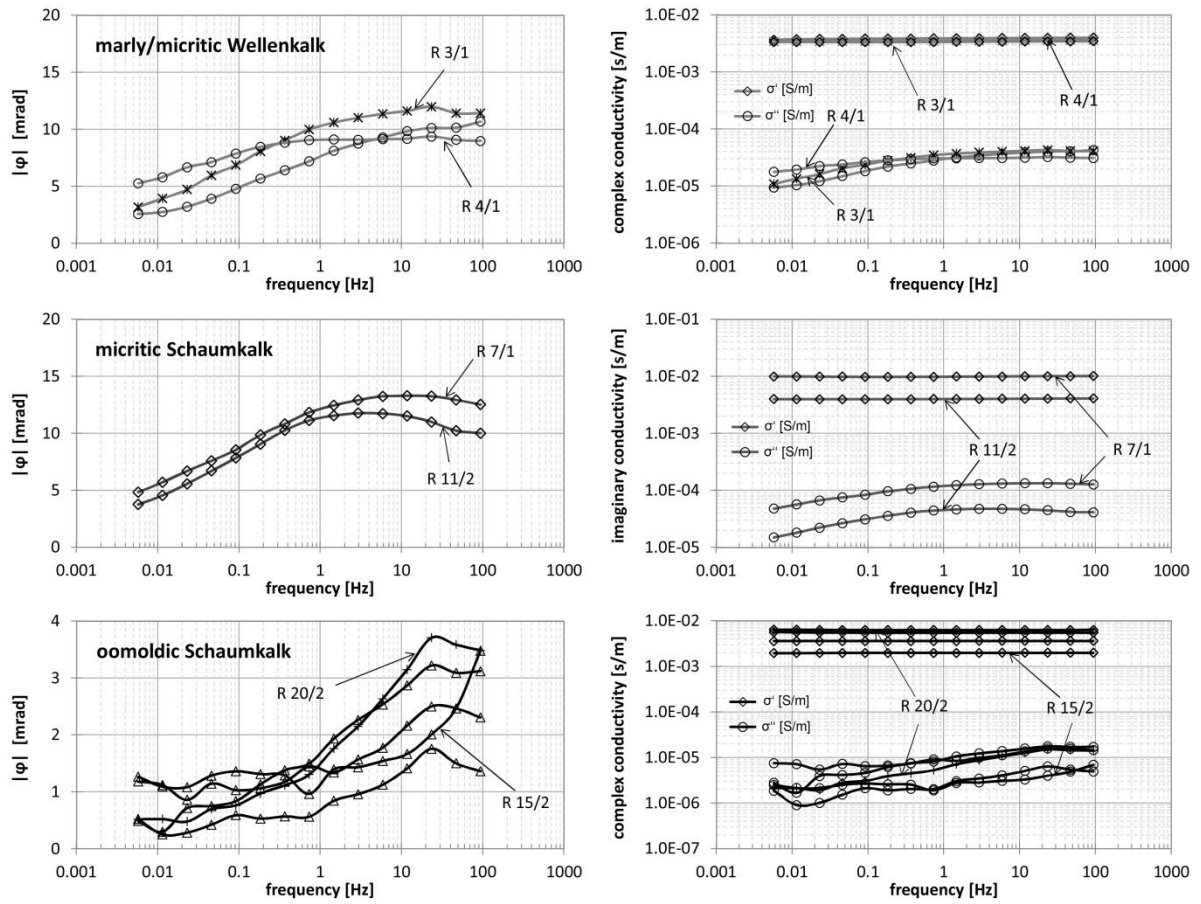


Fig. 5 – SIP phase (φ) and complex conductivity (real part σ' and imaginary part σ'') spectra of marly/micritic Wellenkalk (top), micritic Schaumkalk (center) and oomoldic Schaumkalk (bottom).

To gain deeper insight into the processes that cause relaxation in carbonate rocks, the SIP σ' , σ'' and phase spectra were fitted with Cole-Cole type models as suggested by Pelton et al. (1978) or recently by Loewer et al. (2016). Four parameters are needed for fitting: the inverse of DC specific electrical resistivity R_0 , chargeability m , relaxation time τ and the frequency dependence parameter c . DC resistivity R_0 was derived using a Debye decomposition implementation by Börner et al. (2017) after Nordsiek and Weller (2008). All other parameters were selected to achieve a minimum relative RMS error for the modeled in comparison to the measured data.

The fitting results are exemplified by the diagrams in Fig. 6 which were created for a micritic Wellenkalk sample. They indicate that the real part of complex conductivity is given by the sum of the inverse of DC specific electrical resistivity R_0 and the real part of interface conductivity σ_{int} which presents, however, only a marginal contribution in carbonate rocks.

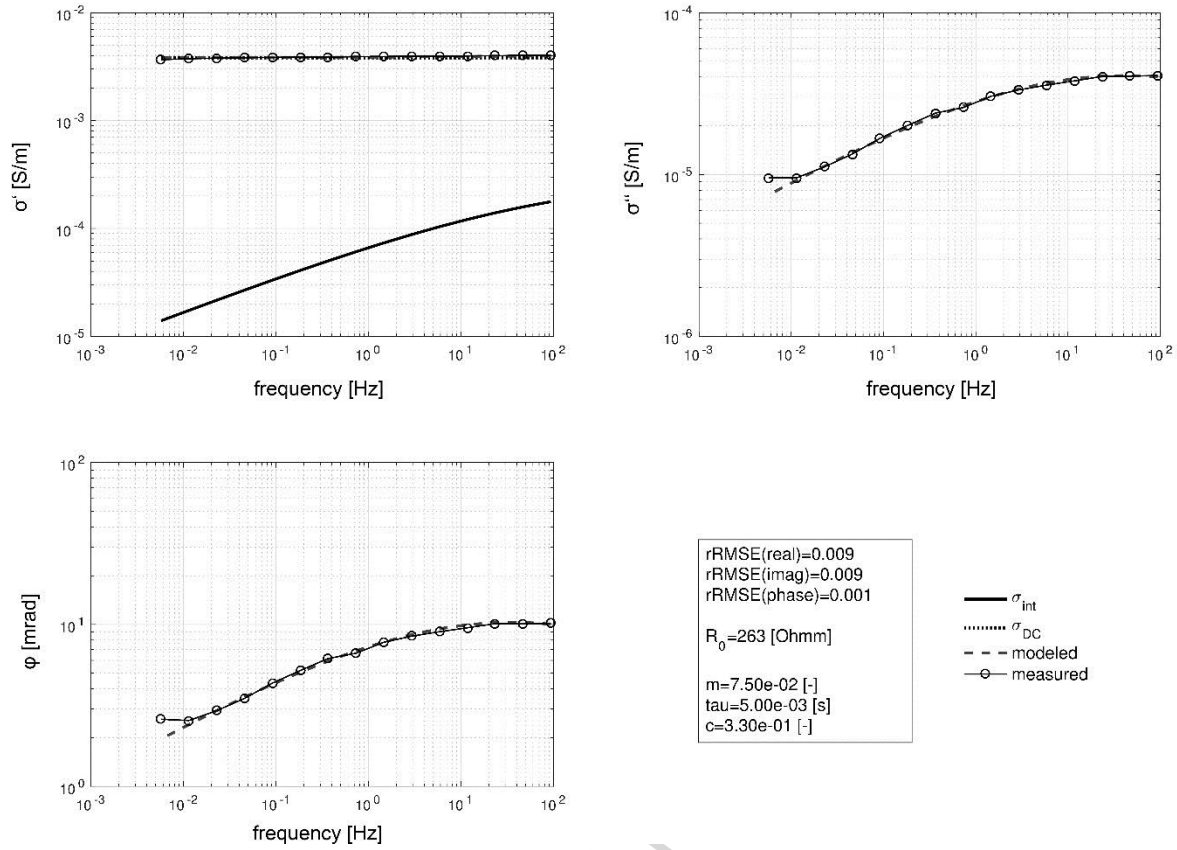


Fig. 6 – Exemplary Cole-Cole fitting results for real (top, left) part σ' , and imaginary (top right) part σ'' of complex conductivity and phase φ (bottom, left) for a micritic Wellenkalk sample whose fabric and pore space are closely similar to sample R 3/1. The box gives relative RMS errors for real part (real) and imaginary part (imag) of complex conductivity and phase, and Cole-Cole parameters R_0 , m , τ (tau), and c . Solid line: real part of interface conductivity σ_{int} ; dotted line: DC conductivity σ_{DC} (the inverse of DC resistivity R_0); dashed line: modeled data; solid line with circles: measured data.

The modeled results suggest that the contribution of interface conductivity to total conductivity is generally low and electrolytic conductivity dominates in all carbonate samples (Table 2). This indicates that the small amounts of pyrite do likely not affect the SIP measurements of the studied samples.

To study the potential influence of pore geometrical effects on SIP-derived parameters, diagrams of selected parameters (complex conductivity, phase φ , chargeability m , frequency dependence c , and relaxation time τ) vs. S_{por} , r_b , r_t and ratio r_b/r_t were created. Fig. 7 shows that the studied samples follow the expected trend of increasing imaginary conductivity with increasing specific surface (Börner and Schön, 1991). Plotting SIP data at 1 Hz is a common convention in petrophysics and ensures the comparability with other studies, thus this frequency was selected for the plots in Fig. 7, bottom.

The real part of complex conductivity reflects a more or less constant level for all samples, while both the imaginary part and phase φ increase with increasing S_{por} . The frequency dependence parameter c exhibits a correlation with S_{por} (Fig. 7) and decreases consistently with increasing S_{por} . This means that the slope of the phase spectrum on a double logarithmic scale increases with decreasing S_{por} . Chargeability m increases with increasing specific surface S_{por} . This indicates that for the grain-dominated oomoldic Schaumkalk samples

which are characterized by low S_{por} , parameter c is comparably high, whereas m is rather low. The mud-dominated fabric of the marly/micritic samples is associated with higher S_{por} and thus parameter c is comparatively lower and chargeability m higher.

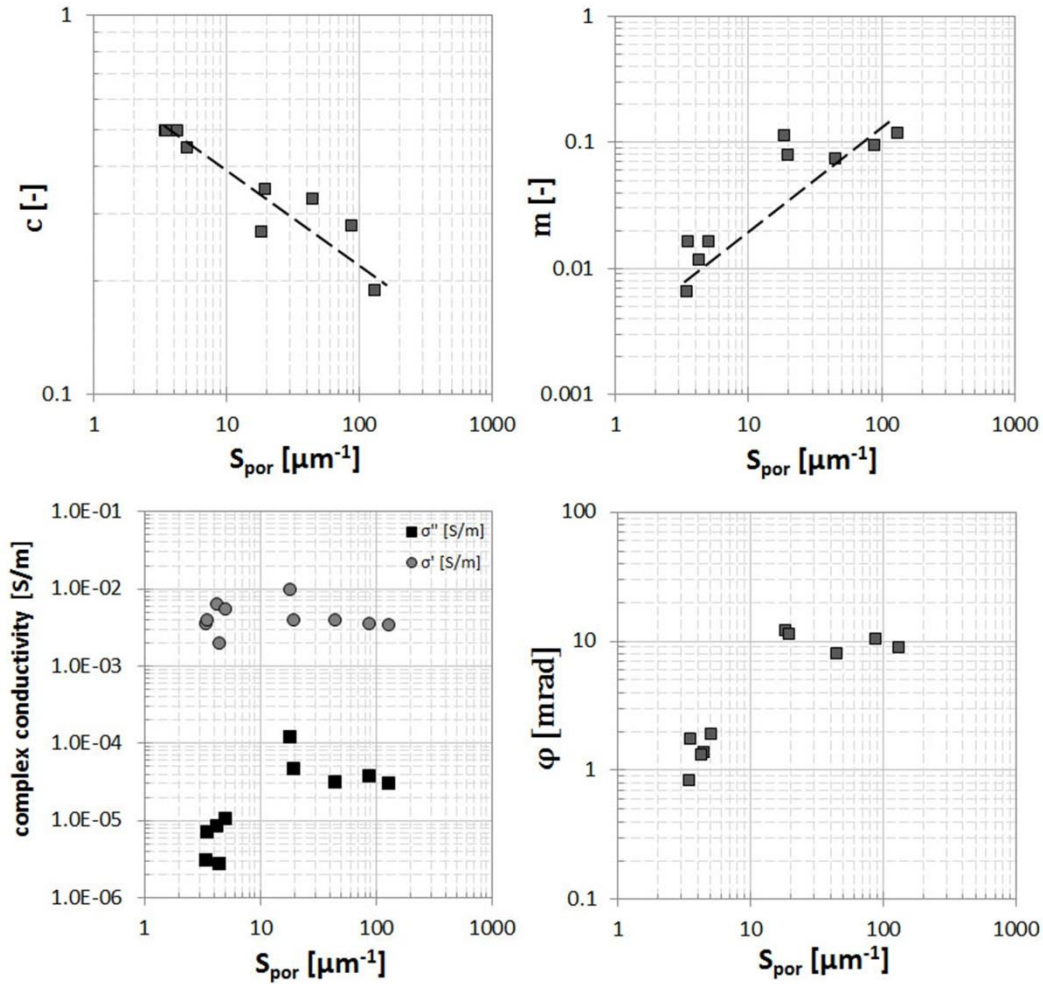


Fig. 7 – Top, left: Frequency dependence parameter c vs. S_{por} [μm^{-1}]. Top, right: Chargeability m vs. S_{por} . Dashed lines were included to highlight the observed trend. Bottom, left: Real (gray dots) part σ' and imaginary (black squares) part σ'' of complex conductivity [S/m] at 1 Hz vs. specific surface S_{por} [μm^{-1}]. Bottom, right: Phase [mrad] at 1 Hz vs. specific surface S_{por} [μm^{-1}].

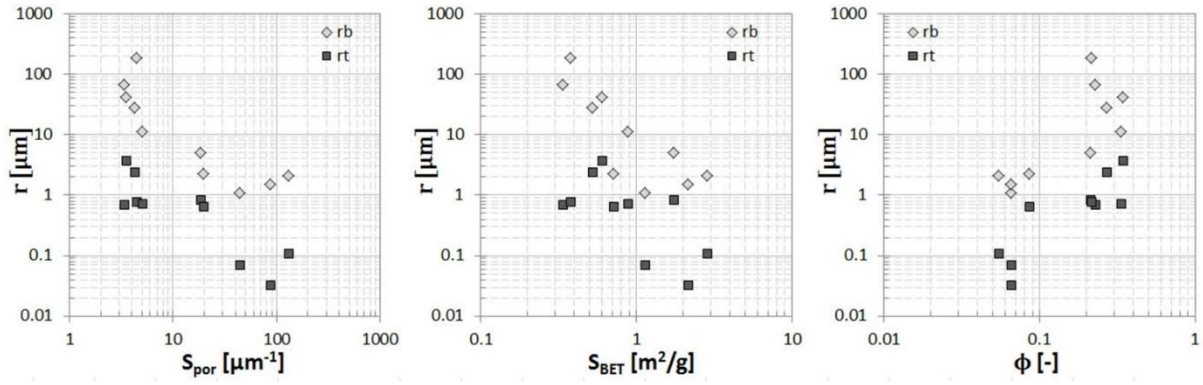


Fig. 8— Left: Pore body radius r_b [μm] and pore throat radius r_t [μm] vs. S_{por} [μm^{-1}]. Center: Pore body radius r_b [μm] and pore throat radius r_t [μm] vs. S_{BET} [$\mu\text{m}^2/\text{g}$]. Right: Pore body radius r_b [μm] and pore throat radius r_t [μm] vs. porosity ϕ [-].

Table 2: DC resistivity R_0 [Ohmm], relaxation time τ [s], chargeability m [-], frequency dependence c [-], dominant NMR or EM pore body radii r_b [μm], dominant MICP pore throat radii r_t [μm] and ratio r_b/r_t of the studied carbonate samples.

Sample	R_0 [Ohmm]	τ [s]	m [-]	c [-]	NMR or SEM r_b [μm]	MICP r_t [μm]	r_b/r_t [-]
R 2/1	263	$5.0 \cdot 10^{-3}$	$7.5 \cdot 10^{-2}$	$3.3 \cdot 10^{-1}$	~1.1	0.071	15
R 3/1	292	$7.5 \cdot 10^{-3}$	$9.75 \cdot 10^{-2}$	$2.8 \cdot 10^{-1}$	~1.5	0.032	46
R 4/1	278	$9.5 \cdot 10^{-3}$	$1.2 \cdot 10^{-1}$	$1.9 \cdot 10^{-1}$	~2.1	0.112	19
R 7/1	104	$1.5 \cdot 10^{-2}$	$1.15 \cdot 10^{-1}$	$2.7 \cdot 10^{-1}$	~5.0	0.838	6.0
R 11/2	256	$3.5 \cdot 10^{-2}$	$8.0 \cdot 10^{-2}$	$3.5 \cdot 10^{-1}$	~2.25	0.662	3.4
R 14/1	277	$5.0 \cdot 10^{-3}$	$6.7 \cdot 10^{-3}$	$5.0 \cdot 10^{-1}$	67.7	0.715	95
R 15/2	507	-	-	-	~190	0.795	239
R 16/1	159	$5.0 \cdot 10^{-3}$	$1.2 \cdot 10^{-2}$	$5.0 \cdot 10^{-1}$	28.0	2.41	12
R 20/2	208	$4.0 \cdot 10^{-3}$	$1.65 \cdot 10^{-2}$	$5.0 \cdot 10^{-1}$	41.6	3.79	11
R 21/1	162	$5.0 \cdot 10^{-3}$	$1.65 \cdot 10^{-2}$	$4.5 \cdot 10^{-1}$	11.3	0.747	15

Fig. 8 (left) shows that a correlation exists between specific surface S_{por} and both pore body radius r_b and pore throat radius r_t . As expected, the smaller both pore throat r_t and pore body radius r_b are, the higher specific surface S_{por} is. This trend can also be observed in a less pronounced manner between S_{BET} and both pore body radius r_b and pore throat radius r_t (Fig. 8, center). Fig. 8 (right) shows that there is also a good correlation between pore body radius r_b and pore throat radius r_t and porosity ϕ . Porosity increases with increasing pore body radius r_b and pore throat radius r_t which highlights that a small number of large pores add more significantly to total porosity than a high number of small pores.

With regard to pore radii, Fig. 9 shows that despite a more or less large scatter in the data, correlations of the Cole-Cole model parameters with both radii r_b and r_t and ratio r_b/r_t can be found. Frequency dependence parameter c probably exhibits a weak correlation with both r_b and r_t and a tendency of increasing with increasing radius. Similar observations can be made for chargeability m where a tendency of decreasing m with increasing radius seems to exist. Time constant τ , in contrast, shows the best correlation with ratio r_b/r_t

whereas no specific relationship with one of the two radii could be found for the investigated carbonate samples.

This first data set implies that the larger either radius r_b or r_t , the higher is parameter c and the lower is m . This behavior results in sharp peaks in the SIP phase and imaginary conductivity spectra with a low chargeability level. This could be a hint that membrane polarization is a relevant mechanism in carbonate rocks, since both pore body and pore throat in combination are leading to a characteristic polarization. Both radii are generally largest in the oomoldic Schaumkalk (Table 2) with its grain-dominated fabric and large, more or less spherical pores. In contrast, the mud-dominated fabric of the marly/micritic Wellenkalk is associated with generally small r_b and r_t resulting in low values for parameter c but comparably higher values for chargeability m . This means that the peaks in the SIP spectra are rather broad and exhibit a high chargeability level. With regard to peak position, the data in this study indicate that it is neither r_b nor r_t that determines its position on the frequency axis, but rather the r_b/r_t ratio. This means that samples with the same r_b/r_t ratio may be characterized by the same peak relaxation time, independently of their absolute pore dimensions, which is the case for, for instance, the micritic sample R 2/1 and the oomoldic sample R 21/1 (Table 2).

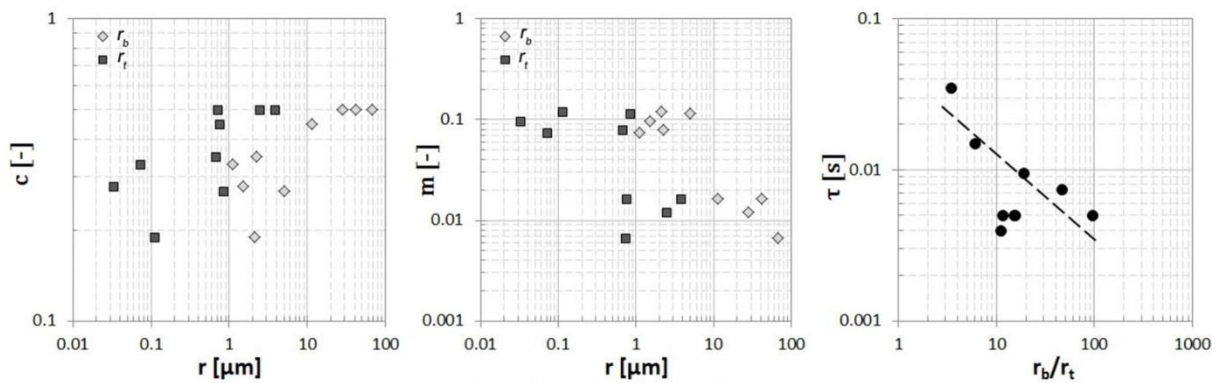


Fig. 9– Left: Frequency dependence parameter c vs. pore radius r [μm]; center: Chargeability m vs. pore radius r [μm]; pore throat radius: black squares, pore body radius: gray rhombs; Right: Time constant τ vs. ratio r_b/r_t .

CONCLUSIONS AND OUTLOOK

In this study, the results of NMR, MICP and SIP measurements on mud-dominated marly/micritic Wellenkalk, micritic Schaumkalk and grain-dominated oomoldic Schaumkalk samples are presented. The findings indicate that the studied carbonate fabric types have distinct pore space features which result in different petrophysical characteristics: The high-porosity, grain-dominated samples are characterized by generally low pore surface area-to-porosity ratios ($3.4\text{--}5.0 \mu\text{m}^{-1}$) and comparatively large dominant r_b ($3.5\text{--}67 \mu\text{m}$) and r_t ($0.7\text{--}3.8 \mu\text{m}$). The low-porosity, mud-dominated samples show generally high specific surface S_{por} ($44\text{--}128 \mu\text{m}^{-1}$), which are associated with low dominant pore body radii r_b (about $1\text{--}2 \mu\text{m}$) and low dominant MICP-derived pore throat radii r_t ($0.03\text{--}0.11 \mu\text{m}$). The calculated r_b/r_t ratios of the grain-dominated samples are, however, not as

different from the mud-dominated samples as could probably be expected (15-46 for micritic samples, 4.5-95 for oomoldic samples).

For mud-dominated carbonates that were deposited under reducing conditions, it is necessary to validate the NMR-derived body radius against other methods because small amounts of pyrite may occur in the samples that affect the T_2 relaxation and lead to prediction of overly small pore sizes. In this case, pore size distributions computed from SEM images delivered valuable information indicating that NMR-derived pore body radii were predicted 1 to 2 orders of magnitude too small. The smallest pore cross sectional area that could be resolved by SEM imaging in this study was $1 \mu\text{m}^2$, which was sufficient considering the resolution provided by MICP measurements. In addition, image-derived pore size distributions are helpful for investigation of the very large pores of the grain-supported oomoldic samples where the largest pores cannot be resolved with NMR measurements as it is difficult to keep them fully saturated. These observations support the suggestion that a combined evaluation of pore space is recommendable to overcome each method's specific resolution and scale limitations. Recent advances in digital rock physics implementing micro and/or nano computer tomography would also be an option to validate the presented pore throat and pore body radius distributions.

Cole-Cole models were used to fit the real and imaginary parts of complex conductivity as well as the measured phase spectra and characteristic parameters (frequency dependence parameter c , chargeability m , relaxation time constant τ) were derived: The larger either r_b or r_t , the higher is parameter c and the lower is m and the larger S_{por} , the lower is c and the higher is m . The grain-dominated samples (large pores, low S_{por}) are therefore characterized by quite sharp peaks in their phase spectra (high c), but low absolute phase values (low m), whereas the mud-dominated samples (small pores, high S_{por}) show broad phase peaks (low c) and comparably high absolute phase values (high m). τ , in contrast, decreases with increasing ratio r_b/r_t and may therefore assume similar values in micritic and oomoldic samples despite their different pore space characteristics. These results present a first step towards the correlation of characteristic SIP parameters with pore geometrical information in carbonate rocks. The correlations that were observed need to be verified by expanding the carbonate data base and including further fabric types. The contribution of minor amounts of pyrite to chargeability in carbonate rocks needs to be studied systematically to correct, for instance, elevated chargeability levels.

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NOMENCLATURE

c	Frequency dependence parameter [-]
f_{peak}	Peak frequency [Hz]
m	Chargeability [-]
p_c	Capillary pressure [Pa]
R_0	DC resistivity [Ohmm]
r	Pore radius [m or μm]
r_b	Pore body radius [μm]
r_c	Critical pore radius [μm]
r_t	Pore throat radius [μm]
S_{BET}	Pore surface area normalized to sample weight [cm^2/g]
S_{por}	Pore -surface-to-pore-volume ratio [μm^{-1}]
S_{pore}	Pore surface [m^2]
S_{tot}	Pore surface area normalized to total sample volume [cm^2/cm^3]
T_2	Transverse NMR relaxation time [ms]
$T_{2,surf}$	Surface relaxation term of T_2 [ms]
V_{pore}	Pore volume [cm^3]
$Z(\omega)$	Complex impedance [Ohmm]
γ	Surface tension [mN/m]
θ	Contact angle [$^\circ$]
ϕ	Porosity [-]
φ	Phase angle [mrad]
ρ_g	Matrix grain density [g/cm^3]
ρ_e	Effective surface relaxivity [$\mu\text{m}/\text{s}$]
ρ_2	Surface relaxivity [$\mu\text{m}/\text{s}$]
σ^*	Complex electrical conductivity [S/m]
σ'	Real part of complex electrical conductivity [S/m]
σ''	Imaginary part of complex electrical conductivity [S/m]
σ_{el}	Electrolytic conductivity [S/m]
σ_{int}^*	Interface conductivity [S/m]
τ	Relaxation time [s]
ω	Angular frequency [Hz]

Acronyms

BET	Brunauer–Emmett–Teller theory
CPMG	Carr-Purcell-Meiboom-Gill sequence
DC	Direct current
FFT	Fast Fourier Transform
MICP	Mercury intrusion
NMR	Magnetic resonance
SIP	Spectral induced polarization
SEM	Scanning electron microscopy
RMS	Root mean square

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Highlights

- High-porosity, grain-dominated carbonates are characterized by generally low pore surface area-to-porosity ratios and comparatively large dominant r_b .
- Low-porosity, mud-dominated samples are characterized by high specific surface S_{por} , low dominant pore body radii r_b , and low dominant MICP-derived pore throat radii r_t .
- The larger either r_b or r_t , the higher is Cole-Cole model parameter c and the lower is m and the larger S_{por} , the lower is c and the higher is m .

ACCEPTED MANUSCRIPT

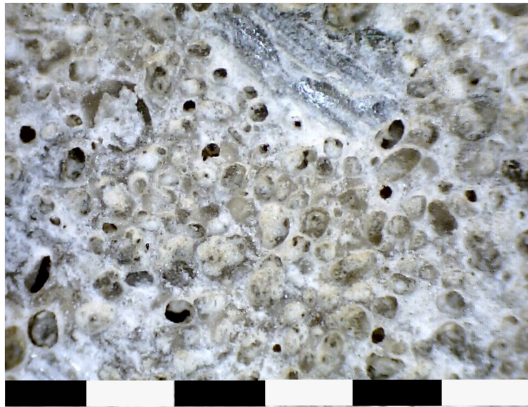
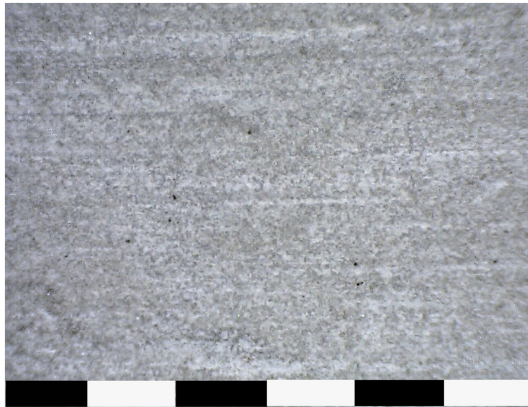


Figure 1

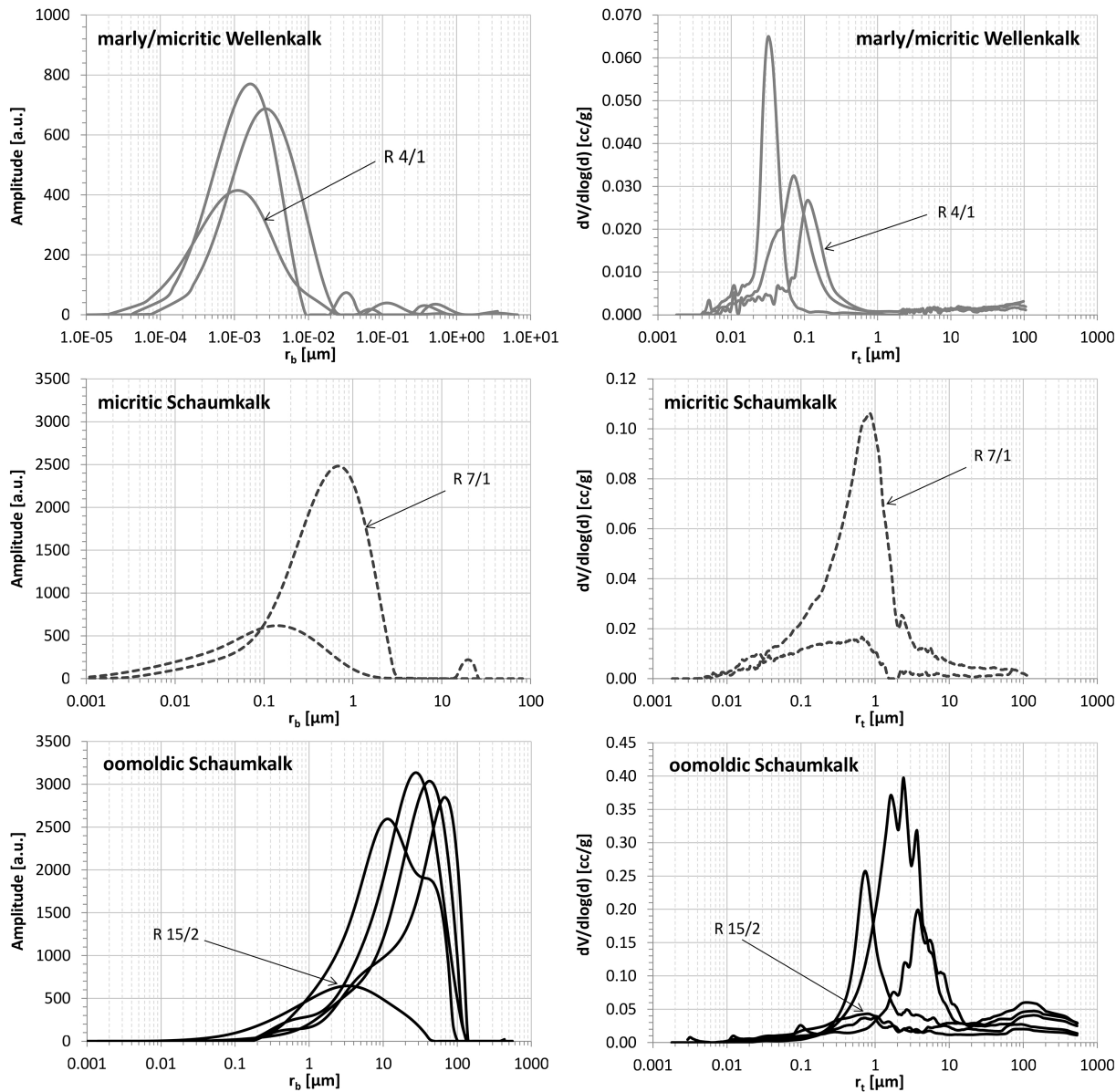


Figure 2

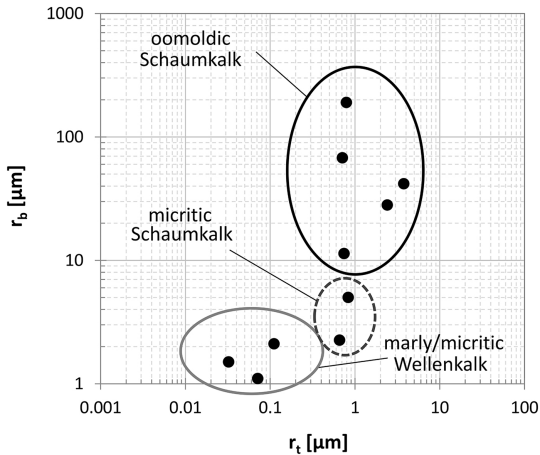


Figure 3

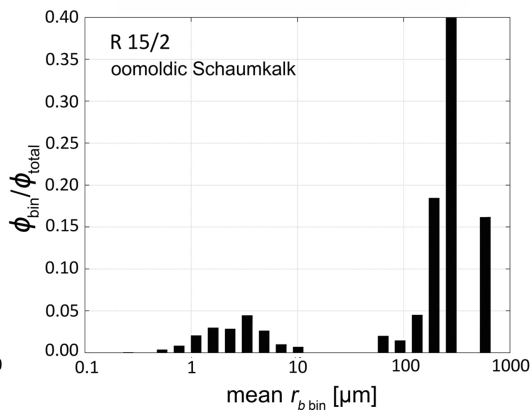
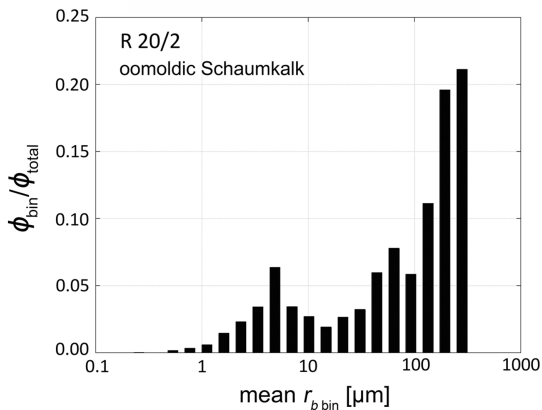
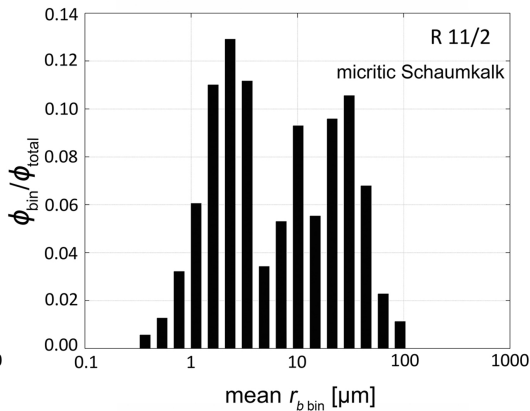
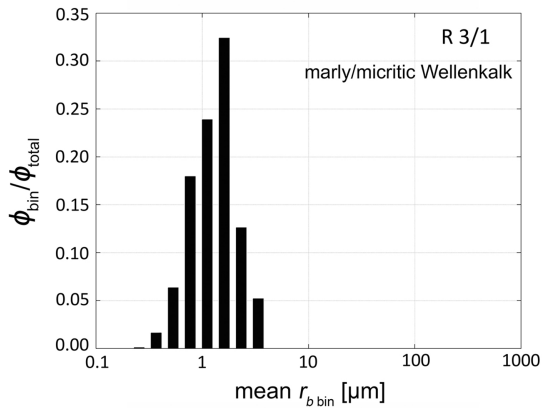


Figure 4

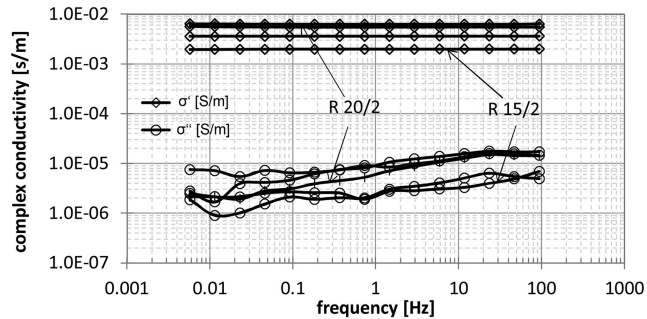
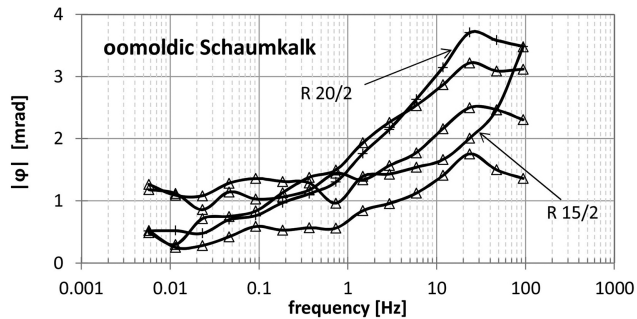
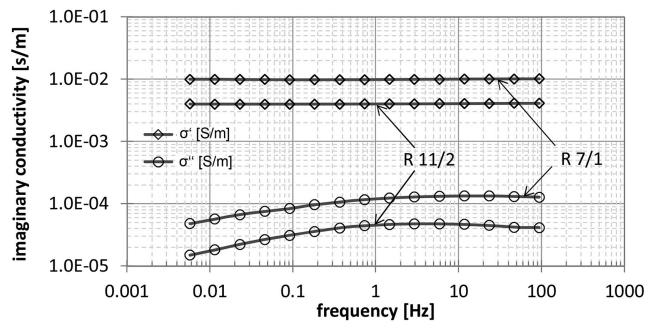
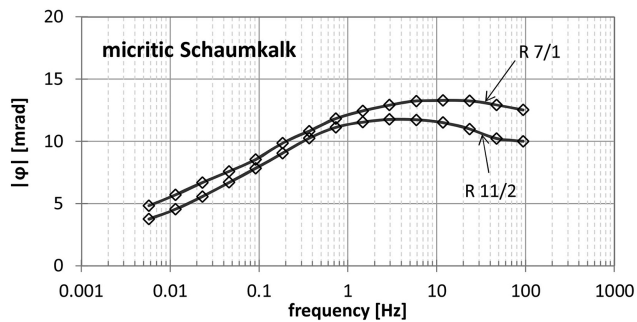
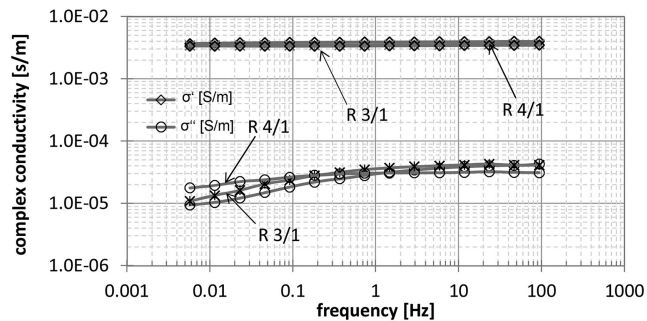
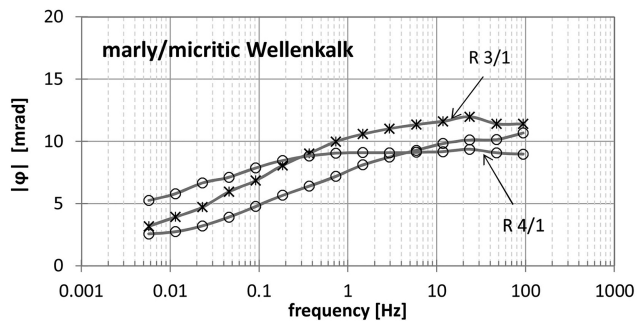
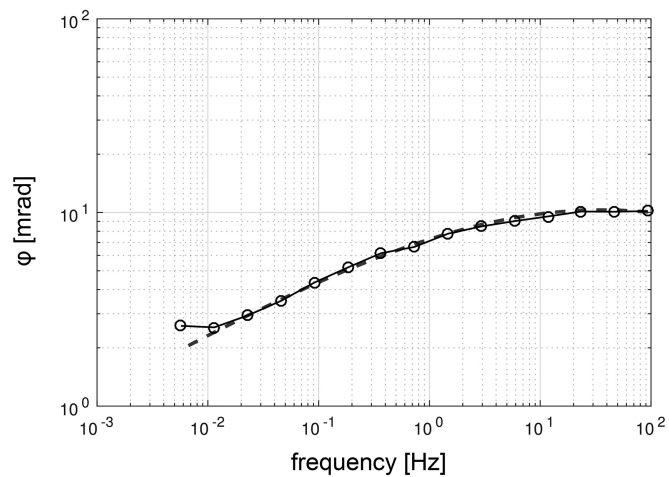
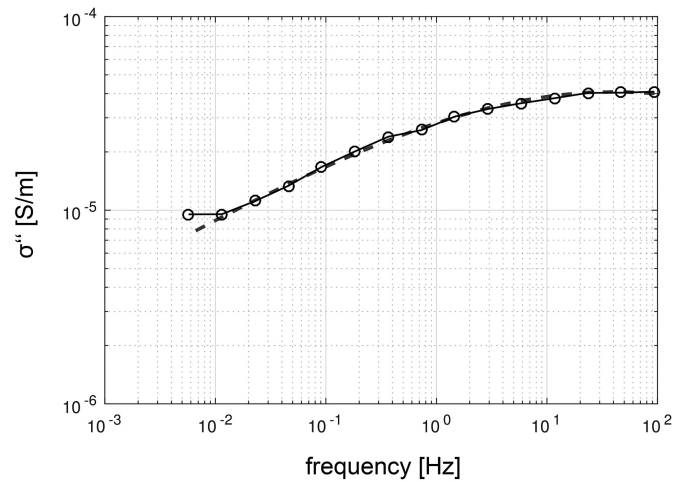
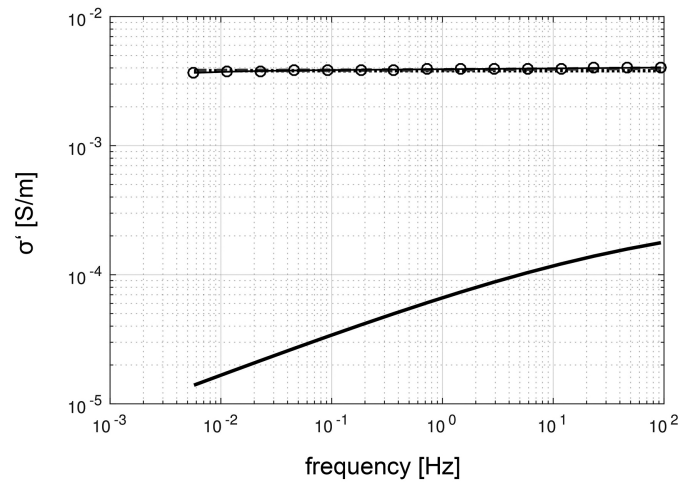


Figure 5



$\text{rRMSE}(\text{real})=0.009$
 $\text{rRMSE}(\text{imag})=0.009$
 $\text{rRMSE}(\text{phase})=0.001$

$R_0=263$ [Ohmm]

$m=7.50\text{e-}02$ [-]

$\tau=5.00\text{e-}03$ [s]

$c=3.30\text{e-}01$ [-]

$\text{—} \sigma_{\text{int}}$
 $\cdots \sigma_{\text{DC}}$
 --- modeled
 $\text{—} \bigcirc \text{ measured}$

Figure 6

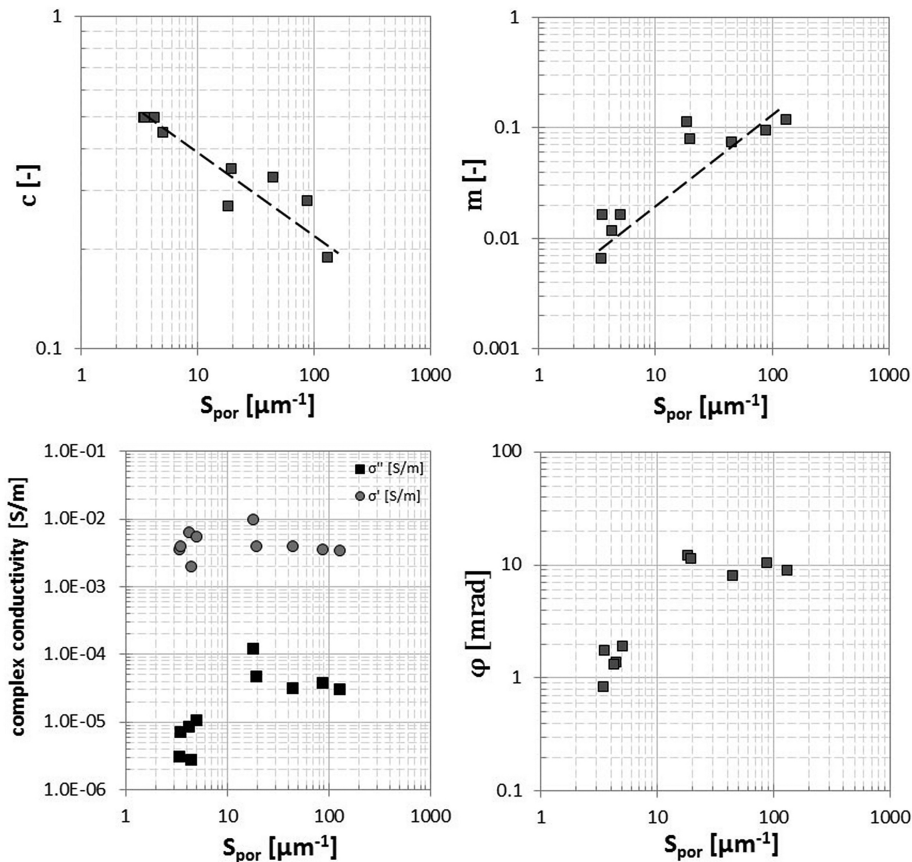


Figure 7

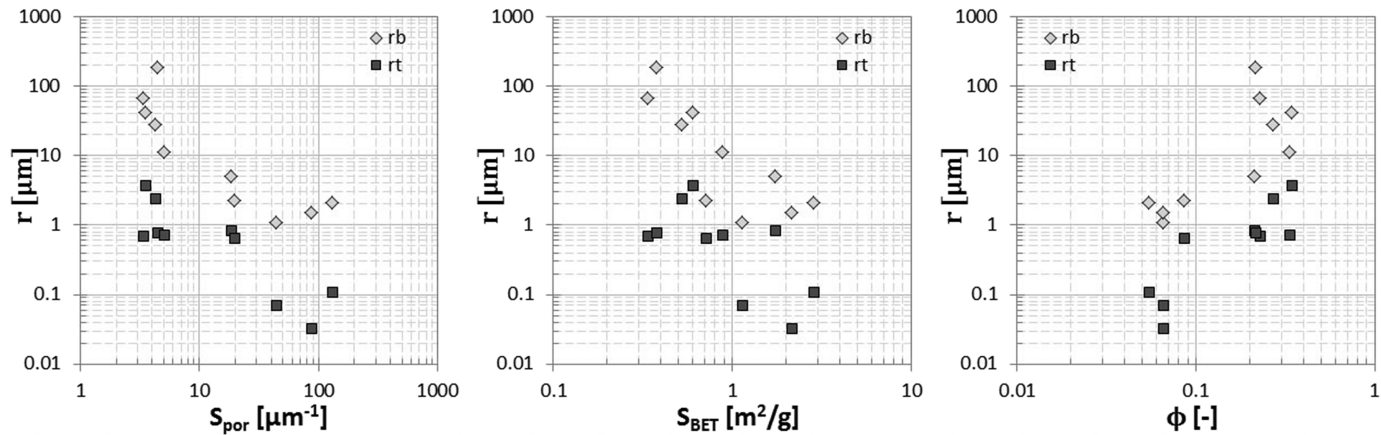


Figure 8

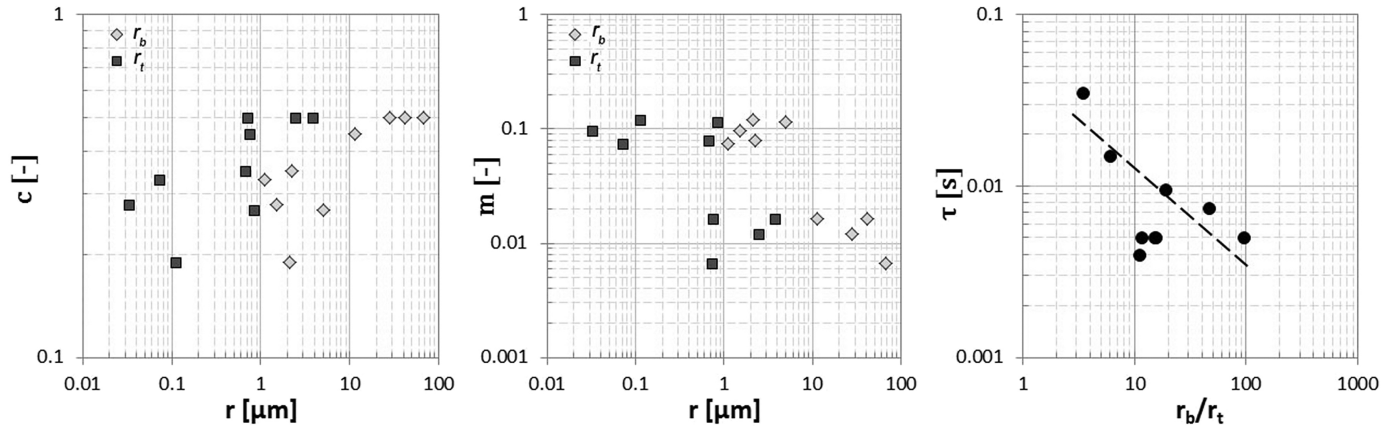


Figure 9